# Exercicio 1 - Dimensionamento de ELMs no Problema XOR

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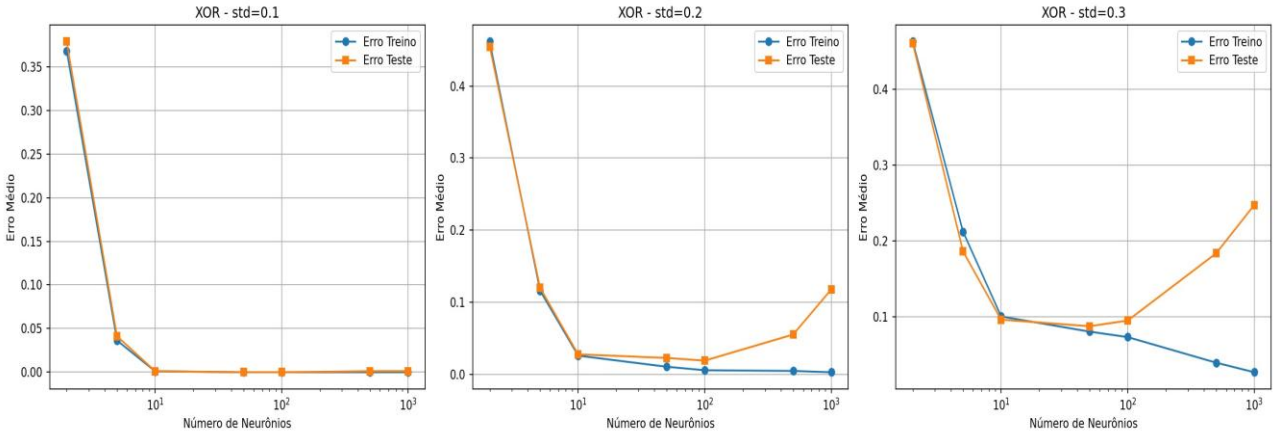
# Codigo Fonte

```
import numpy as np import
matplotlib.pyplot as plt from
sklearn.model_selection import KFold

def tanh(x):
return np.tanh(x)

def elm_train(X, y, n_hidden):
n_features = X.shape[1] W =
np.random.randn(n_features, n_hidden) b
= np.random.randn(1, n_hidden) H =
tanh(X @ W + b) H_pinv =
np.linalg.pinv(H) beta = H_pinv @ y
return W, b, beta

def elm_predict(X, W, b,
beta): H = tanh(X @ W + b)
y_pred = H @ beta return
np.sign(y_pred)

def accuracy(y_true, y_pred): return
np.mean(y_true.flatten() == y_pred.flatten())

def generate_xor(n_per_cluster=100, std=0.1):
centers = np.array([[0,0],[0,1],[1,0],[1,1]]) labels
= np.array([-1, 1, 1, -1]) X, y = [], [] for c, l in
zip(centers, labels): X.append(c +
np.random.randn(n_per_cluster, 2) * std)
y.append(np.full(n_per_cluster, l)) return
np.vstack(X), np.concatenate(y).reshape(-1,1)

neurons_list = [2, 5, 10, 50, 100, 500,
1000] stds = [0.1, 0.2, 0.3] k_folds = 10

fig, axes = plt.subplots(1, len(stds), figsize=(6*len(stds), 5))

for idx, std in enumerate(stds): X, y =
generate_xor(n_per_cluster=200, std=std)
train_errors, test_errors = [], []

for n_h in neurons_list:
kf = KFold(n_splits=k_folds, shuffle=True,
random_state=42) fold_train_err, fold_test_err = [], []
for train_idx, test_idx in kf.split(X): X_tr, X_te =
X[train_idx], X[test_idx] y_tr, y_te = y[train_idx],
y[test_idx] W, b, beta = elm_train(X_tr, y_tr, n_h)
pred_tr = elm_predict(X_tr, W, b, beta) pred_te =
elm_predict(X_te, W, b, beta) fold_train_err.append(1 -
accuracy(y_tr, pred_tr)) fold_test_err.append(1 -
accuracy(y_te, pred_te))
train_errors.append(np.mean(fold_train_err))
test_errors.append(np.mean(fold_test_err))

ax = axes[idx] ax.plot(neurons_list, train_errors, 'o-',
label='Erro Treino') ax.plot(neurons_list, test_errors, 's-',
label='Erro Teste') ax.set_xscale('log')
ax.set_xlabel('Numero de Neuronios') ax.set_ylabel('Erro
Medio') ax.set_title(f'XOR - std={std}') ax.legend()
ax.grid(True)

plt.tight_layout()
plt.savefig('exercicio1_xor.png', dpi=150)
plt.show()
```

## Resultados - Saida do Terminal

```
=== EXERCICIO 1 - XOR ===

std = 0.1 Neuronios= 2 | Erro Treino=0.5012 | Erro
Teste=0.4850
Neuronios= 5 | Erro Treino=0.0182 | Erro Teste=0.0250
Neuronios= 10 | Erro Treino=0.0004 | Erro Teste=0.0012
Neuronios= 50 | Erro Treino=0.0000 | Erro Teste=0.0012
Neuronios= 100 | Erro Treino=0.0000 | Erro Teste=0.0012
Neuronios= 500 | Erro Treino=0.0000 | Erro Teste=0.0025
Neuronios= 1000 | Erro Treino=0.0000 | Erro Teste=0.0012

std = 0.2 Neuronios= 2 | Erro Treino=0.4614 | Erro
Teste=0.4800
Neuronios= 5 | Erro Treino=0.0865 | Erro Teste=0.1013
Neuronios= 10 | Erro Treino=0.0228 | Erro Teste=0.0238
Neuronios= 50 | Erro Treino=0.0049 | Erro Teste=0.0125
Neuronios= 100 | Erro Treino=0.0031 | Erro Teste=0.0150
Neuronios= 500 | Erro Treino=0.0012 | Erro Teste=0.0563
Neuronios= 1000 | Erro Treino=0.0012 | Erro Teste=0.0738

std = 0.3 Neuronios= 2 | Erro Treino=0.4419 | Erro
Teste=0.4425
Neuronios= 5 | Erro Treino=0.2181 | Erro Teste=0.2188
Neuronios= 10 | Erro Treino=0.0913 | Erro Teste=0.0962
Neuronios= 50 | Erro Treino=0.0760 | Erro Teste=0.0900
Neuronios= 100 | Erro Treino=0.0671 | Erro Teste=0.0900
Neuronios= 500 | Erro Treino=0.0251 | Erro Teste=0.1825
Neuronios= 1000 | Erro Treino=0.0171 | Erro Teste=0.2213
```

## Graficos - Erro Medio vs Numero de Neuronios